

Module 1: Introduction to Artificial Neural Networks (ANNs)

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Wadhvani School of AI , IIT Madras

Getting Ready for the Course

Knowledge Prerequisites

- **Machine Learning:** Solid understanding of **Linear & Logistic Regression**, and Multiclass Classification.

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- **Essential Libraries:**
 - The core frameworks: **PyTorch** or **TensorFlow**.
 - Scientific computing and plotting: **NumPy** & **Matplotlib**.

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- **Development Environment:**
 - Highly recommended: **Jupyter Notebooks** or **Google Colab**.

Outline of this Module

- 1 From Machine Learning to Deep Learning
- 2 The Anatomy of a Neural Network
- 3 The Learning Loop: How Neural Networks Learn

Machine Learning Landscape

From Traditional Machine Learning to Deep Learning

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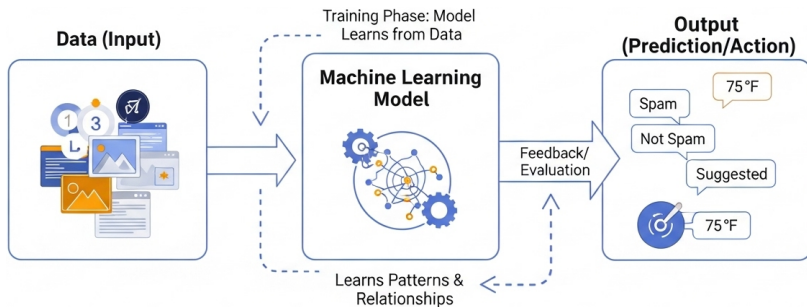


Figure: ML: Learning from Data (Source: ImageGen)

Common Machine Learning Applications



Figure: Housing Price Prediction

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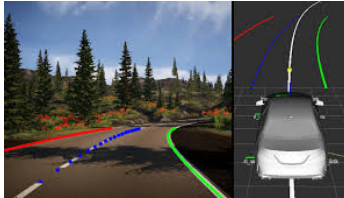


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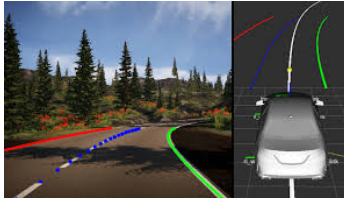


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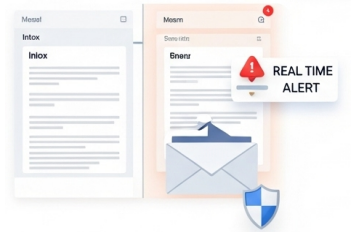


Figure: Message Sentiment Analysis

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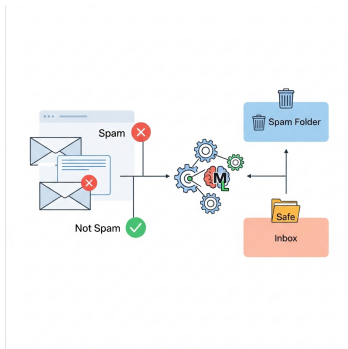


Figure: Spam Email Filtering

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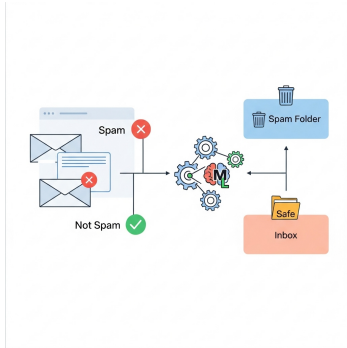


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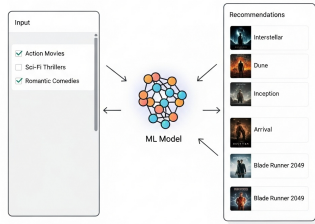


Figure: Recommendation engines

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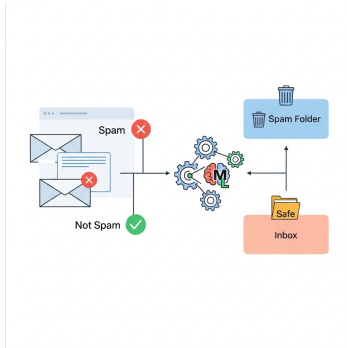


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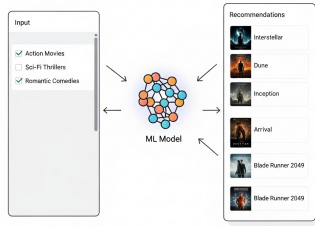


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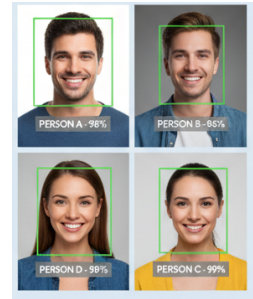


Figure: Image Recognition

The Data & Process

Machine Learning: Key Concepts

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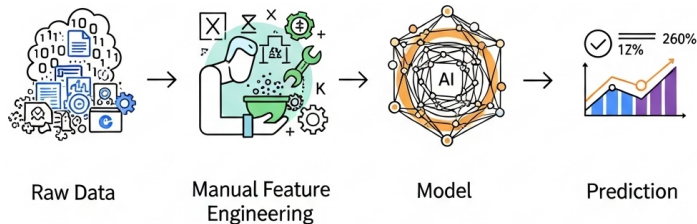


Figure: Feature Engineering Bottleneck
(Source: ImageGen)

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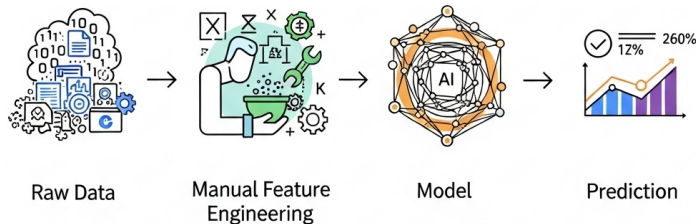


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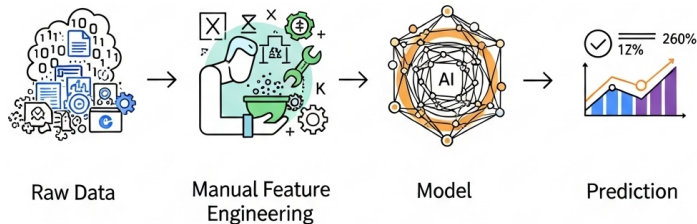


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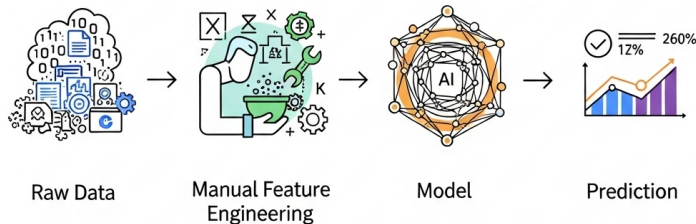


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- **Performance plateaus:** Adding more data yields diminishing returns after a certain point.

The Deep Learning Revolution

From Traditional Machine Learning to Deep Learning

The Shift to Deep Learning: A New Paradigm

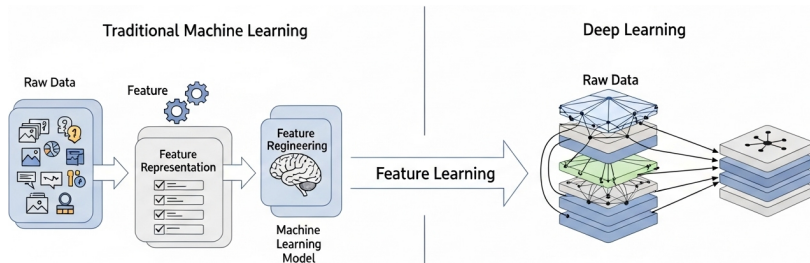


Figure: ML needs manual feature extraction; DL learns them automatically.

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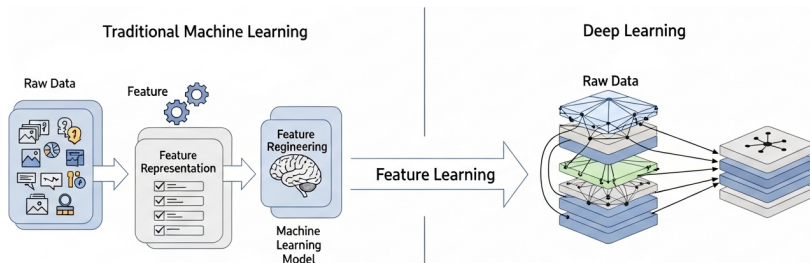


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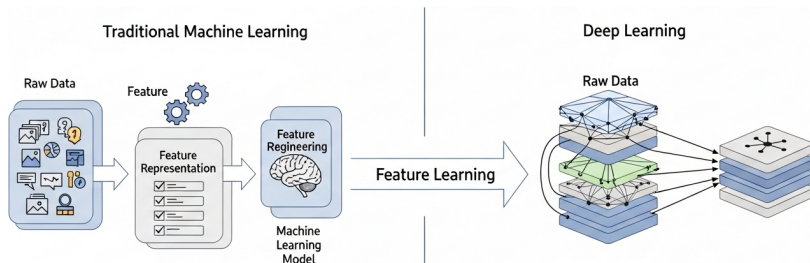


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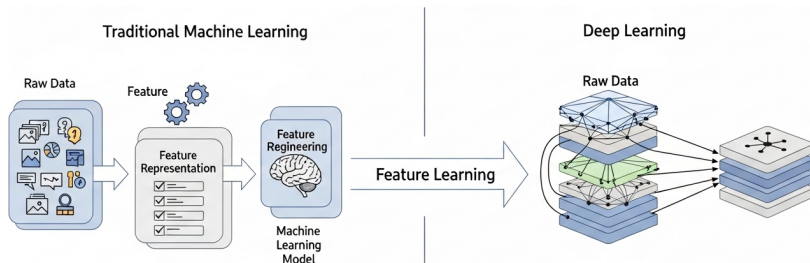


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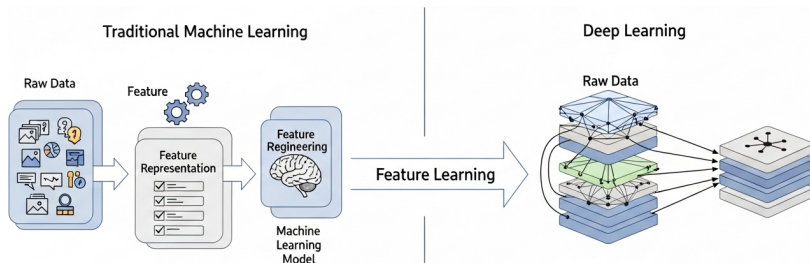


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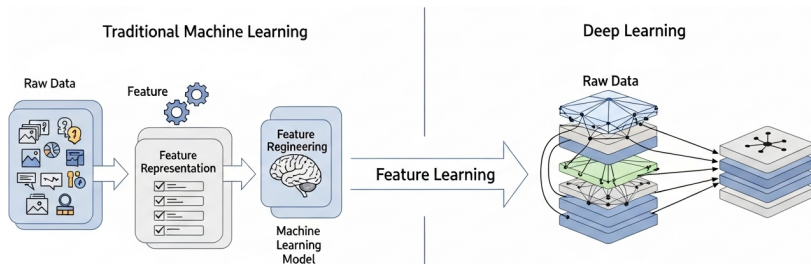


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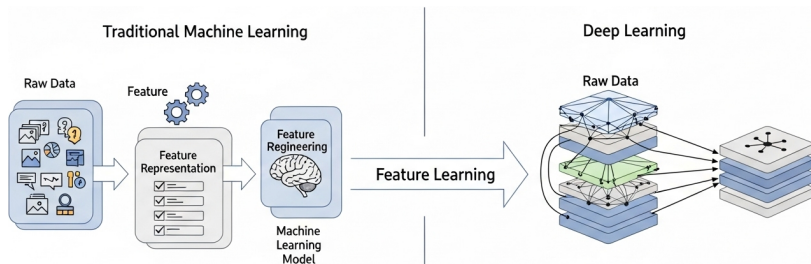


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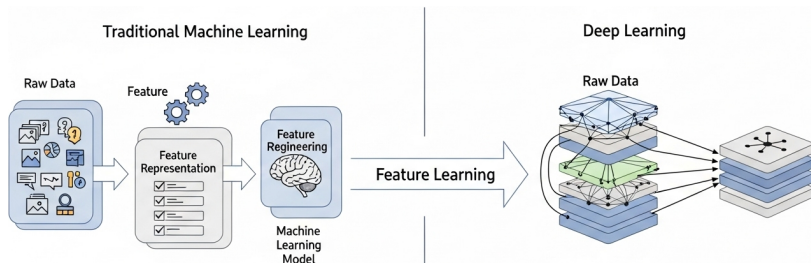


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- Unlocks new, complex capabilities.

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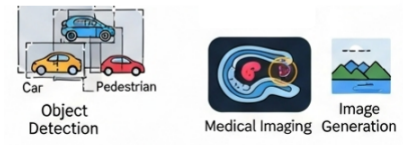


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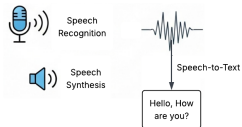


Figure: Speech recognition and Audio synthesis

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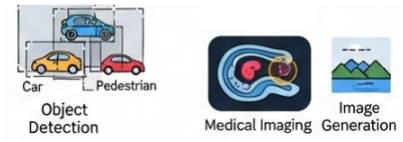


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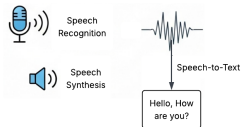


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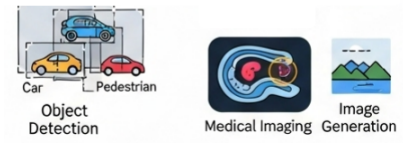


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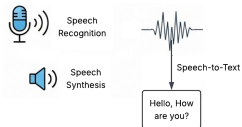


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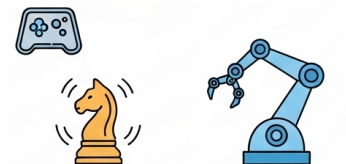


Figure: Game playing and robotics

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- **Training Paradigm:**
Enables **end-to-end** learning.

What Enabled the Deep Learning Revolution?

A convergence of three key factors created the "perfect storm"

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Big Data

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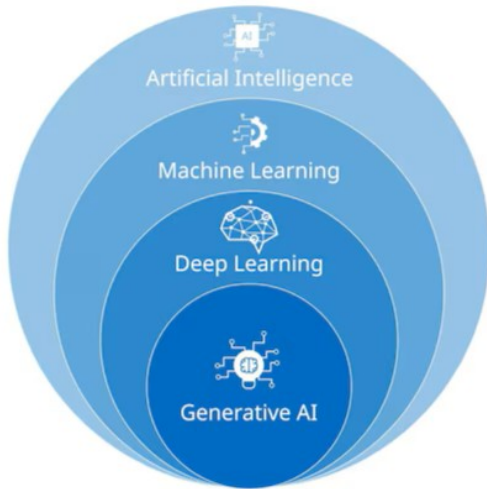
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Better Algorithms

- Architectural innovations.
- e.g., CNNs, Transformers.
- Smarter training methods.

The AI Venn Diagram



From Biology to Mathematics

The Anatomy of a Neural Network

Inspiration from the Brain

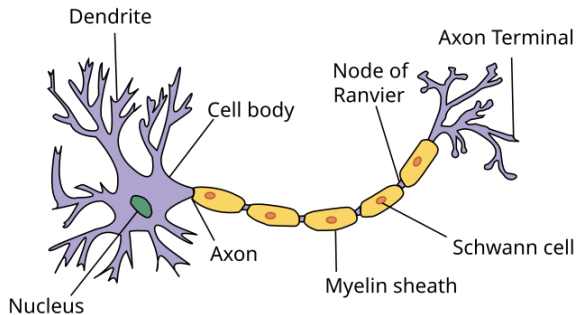


Figure: A biological neuron (Source: Wikipedia)

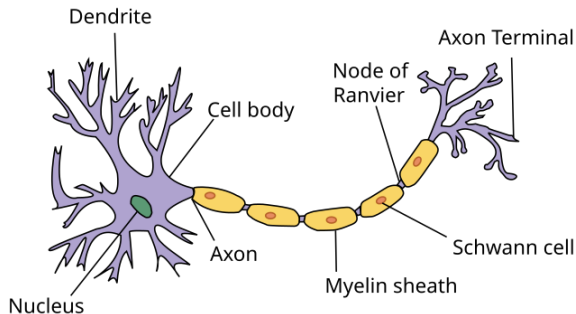


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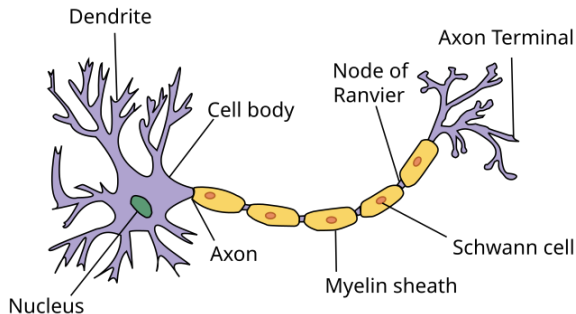


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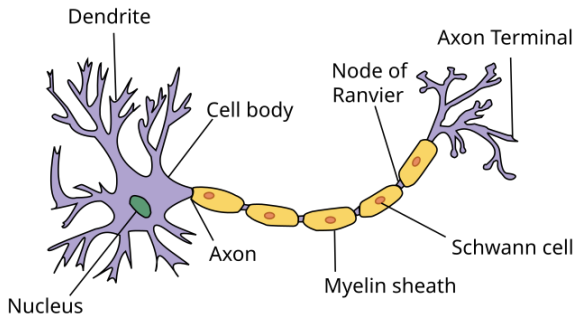


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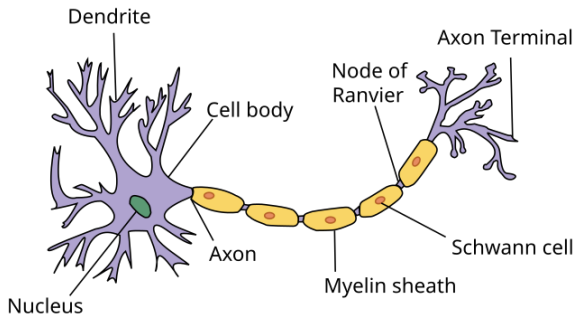


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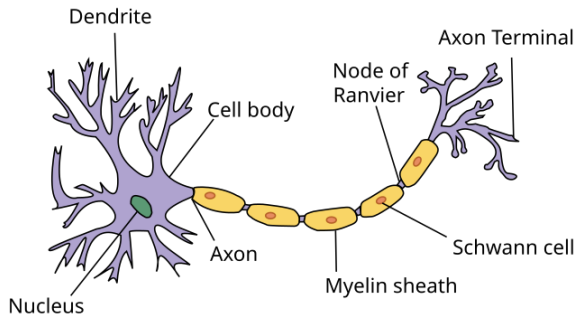


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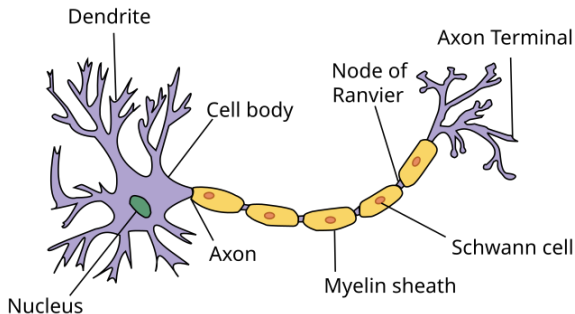


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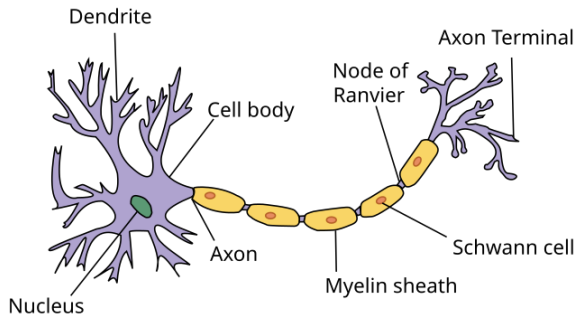


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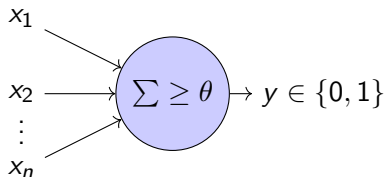


Figure: The M-P Neuron Model: Sum inputs and compare to a threshold θ .

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How it Works:

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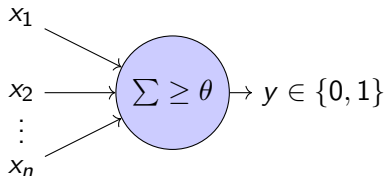


Figure: The M-P Neuron Model: Sum inputs and compare to a threshold θ .

The McCulloch-Pitts Neuron (1943)

The Core Idea: A Simple Biological Logic Gate

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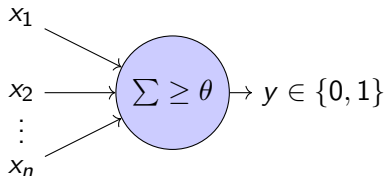


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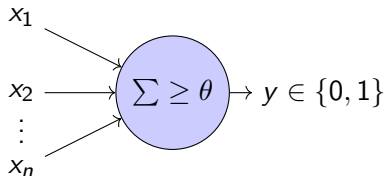


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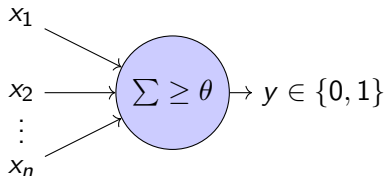


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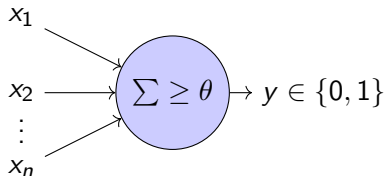


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Significance:

- Proved that networks of these simple units could compute any logical function (e.g., AND, OR).

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Restricted to Binary I/O

- Inputs must be 0 or 1.
- Outputs are only 0 or 1.
- Can't handle **real-valued data**.

The Leap to Learning: The Perceptron (1958)

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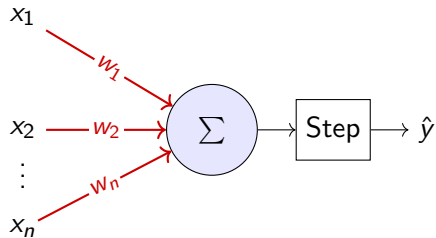


Figure: The Perceptron: Introduces learnable weights (w_i) on each input.

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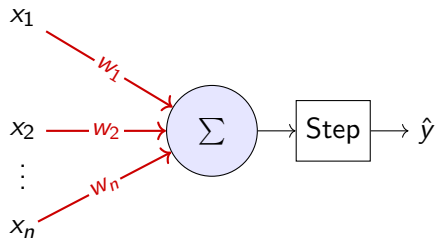


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- Each input is multiplied by a **weight**.
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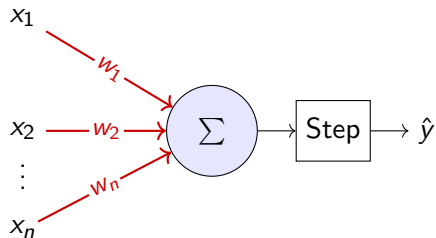


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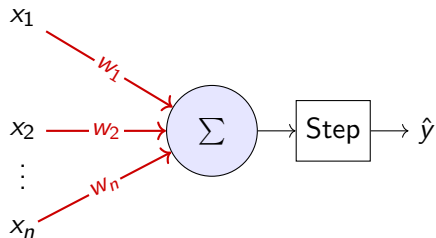


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Significance: The Birth of Learning Machines

For the first time, **a machine could automatically learn** to classify patterns, laying the foundation for all of modern deep learning.

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- **Harsh Threshold:** The step function activation is not differentiable, preventing modern gradient-based training.

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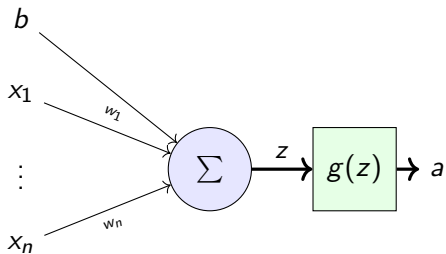


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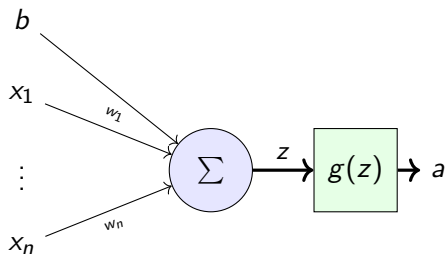


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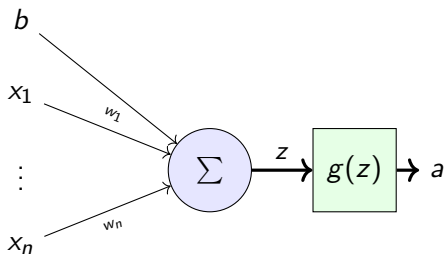


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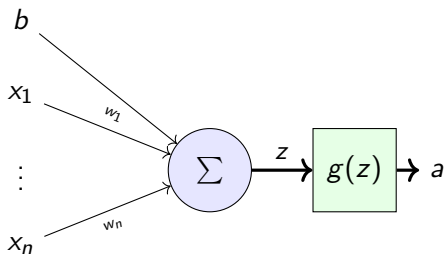


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Why Differentiability is the Superpower?

Differentiable $\rightarrow$ **gradient**. This gradient is used to adjust weights to fix errors. It is the core requirement for **backpropagation**.

Fundamental Tricks in Deep Learning

The Anatomy of a Neural Network

The Numerization Trick: From Sensation to Numbers

The Core Problem: Computers Don't See, They Calculate

- Humans perceive the world through qualitative sensations.

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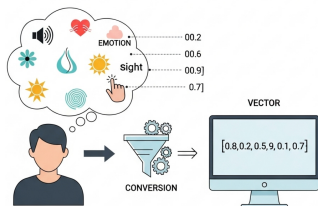
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Our Goal

Convert every piece of information about a problem into a list of numbers, called a **vector**.

Figure: Qualitative representation to Quantitative representation

The Numerization Trick: Example

Scenario: A Self-Driving Car

A car's AI must perceive the road, identify objects, and decide how to act in real-time. **Task:** How do we represent its world as Input and Output vectors?

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- **Object Predictions:** For each detected object, a vector like '[Class, X-Pos, Y-Pos, Velocity]'.

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E.g., ChatGPT can be seen as one, massive function taking your prompt as input.

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The Model as a Function

We express the model's prediction as

$$\hat{y} = f(x; \theta),$$

where:

- x is the numerical input data.
- f is the network architecture (e.g., layers, activation functions).
- θ represents all the weights and biases in the network.
- $\hat{y}$ is the model's prediction.

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- Our goal is to find the parameters θ^* that result in the **lowest possible loss**.

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 $\mathcal{L}(y, \hat{y}) = \mathcal{L}(y, f(x; \theta)) \approx \mathcal{L}(\theta)$, scores how "bad" our model is for a given set of parameters θ .
- High Loss = Bad Model.
Low Loss = Good Model.

Step 2: The Goal

- Our goal is to find the parameters θ^* that result in the **lowest possible loss**.

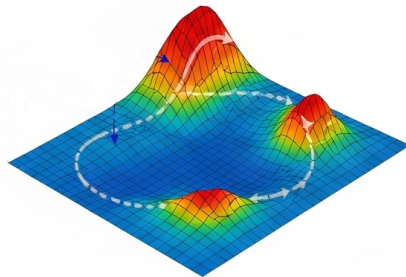


Figure: This creates a high-dimensional **Loss Landscape** with mountains (high error) and valleys (low error).

Feedforward Network Architecture

The Anatomy of a Neural Network

Network Topology: Building a Feedforward Network

1. The Input Layer

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- Receives the data and just pass values along.

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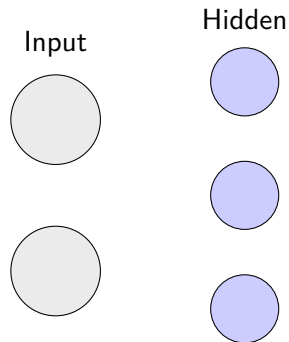
Network Topology: Building a Feedforward Network

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- The network's computational engine.
- They learn increasingly abstract features.



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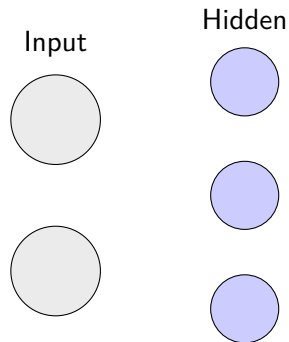
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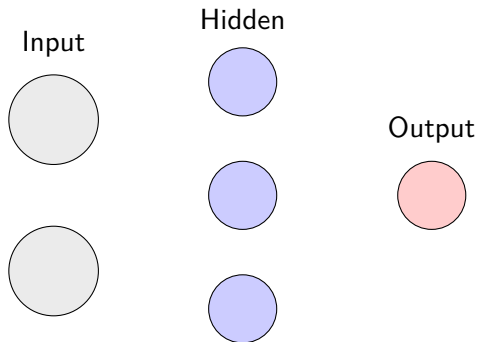
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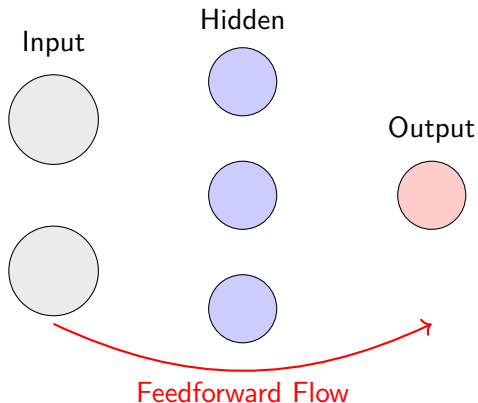
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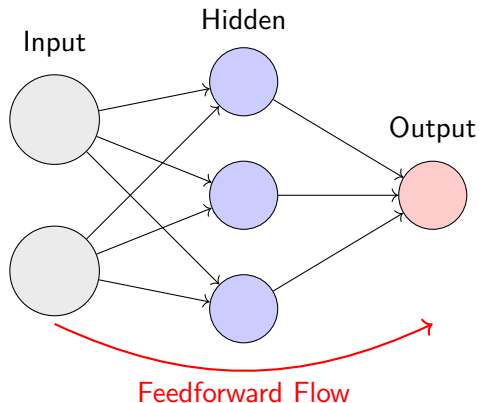
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- **Fully Connected:** Every neuron connects to every neuron in the *next* layer.



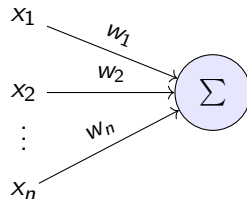
The Standard Forward Model: A Neuron's Calculation

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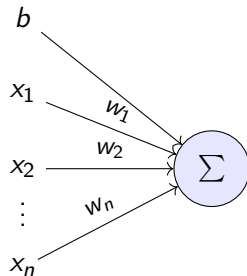
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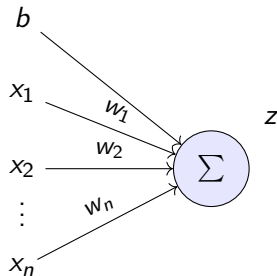
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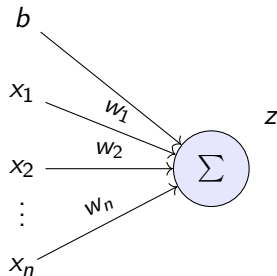


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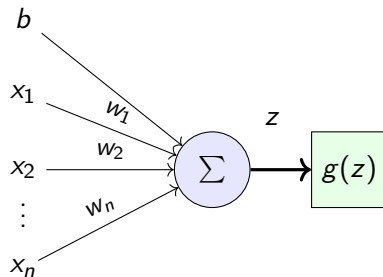
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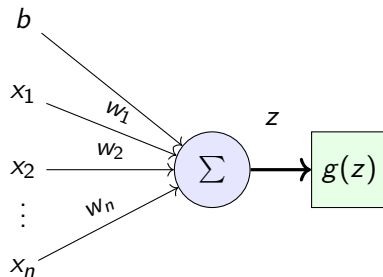
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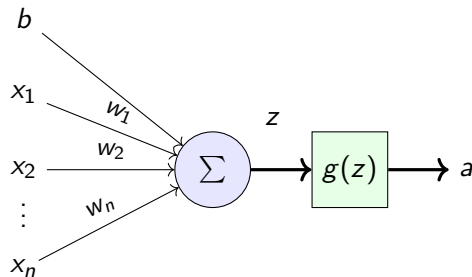
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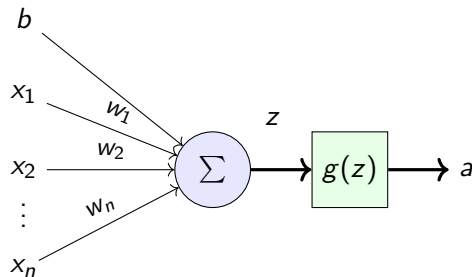
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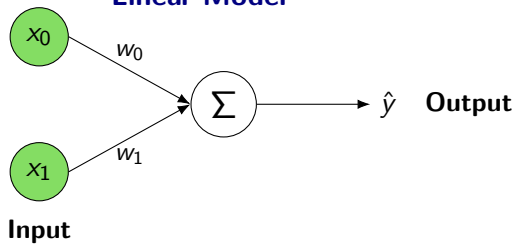


The Fundamental Building Block

This two-step 'Linear Sum $\rightarrow$ Non-Linear Activation' process is performed by every single neuron in a deep neural network.

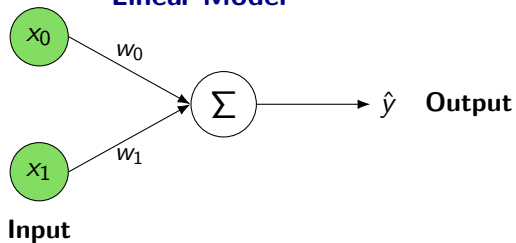
Linear Model vs Artificial Neural Networks

Linear Model



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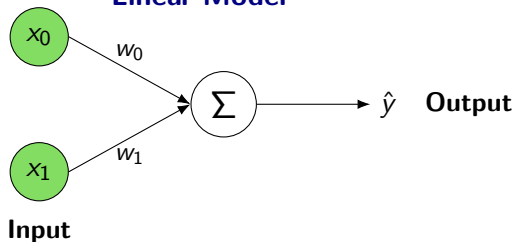
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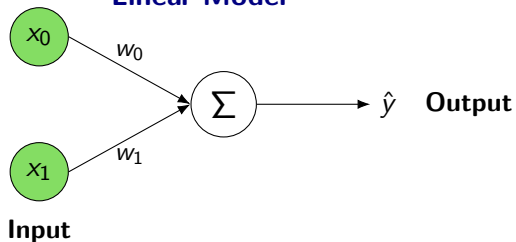


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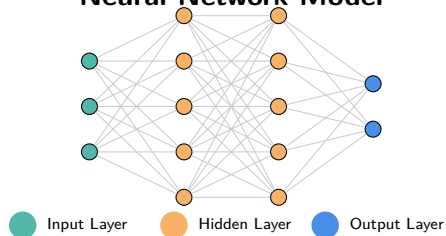
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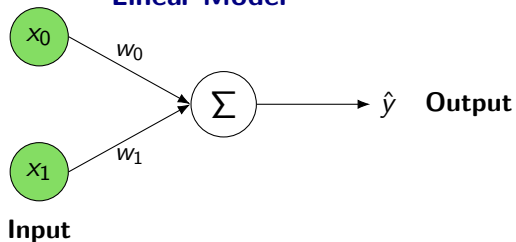
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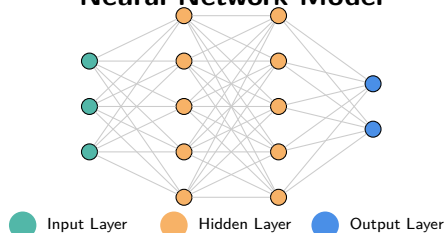
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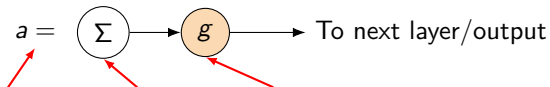
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Neural Network Model



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A **Neuron** has a **linear** and a **nonlinear** operation

The Forward Pass: The Vectorized Layer View

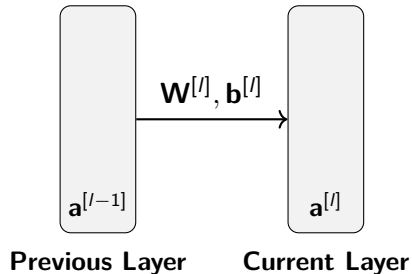
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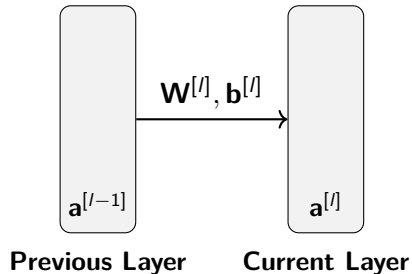
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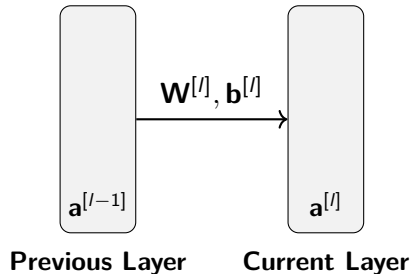
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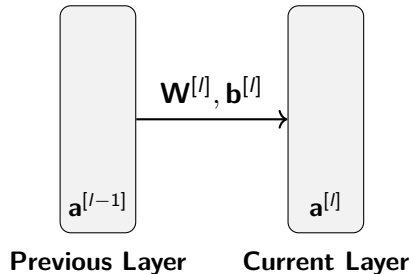
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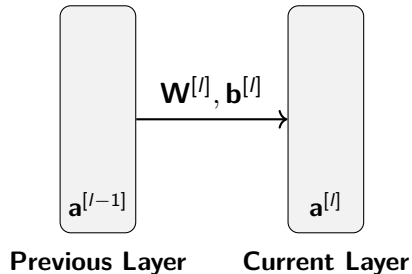
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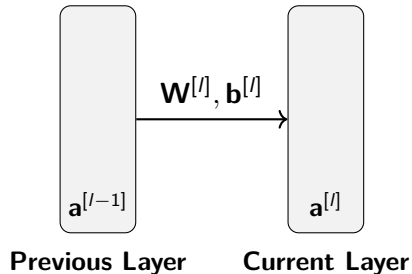
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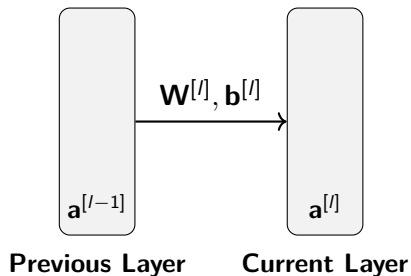
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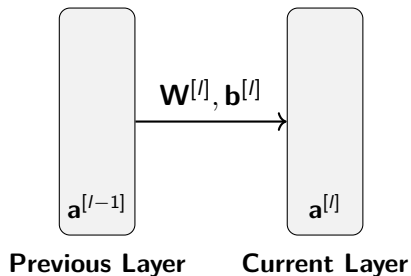
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The Two Core Equations (Vectorized)

Figure: A layer's computation maps an input vector to an output vector.

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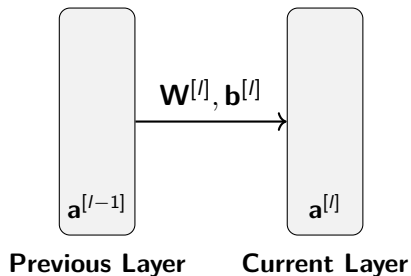


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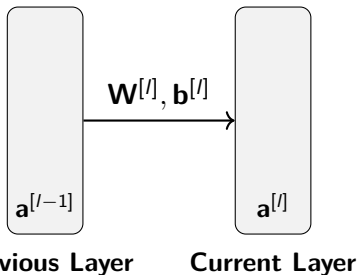


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This is computationally efficient and perfectly suited for GPUs.

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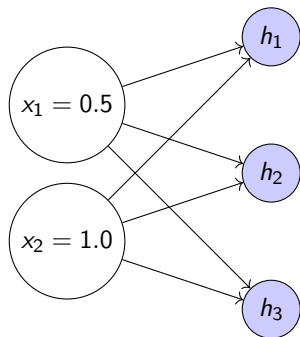


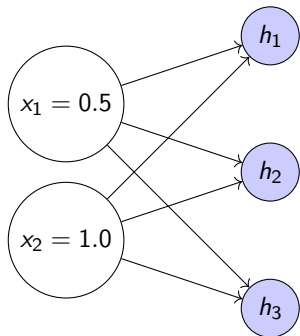
Figure: Input vector $\mathbf{x}$ has 2 features.
Hidden layer has 3 neurons.

Let the activation function be ReLU

$$g(z) = \max(0, z)$$

Let's compute the output of a hidden layer with 3 neurons, receiving input from 2 neurons.

1. Define inputs, weights, and biases:



$$\mathbf{x} = \begin{bmatrix} 0.5 \\ 1.0 \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} 0.2 & 0.7 \\ -0.4 & 0.1 \\ 0.9 & -0.3 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 0.1 \\ 0.2 \\ -0.5 \end{bmatrix}$$

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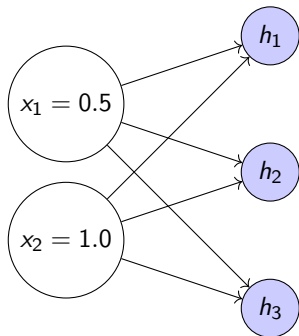


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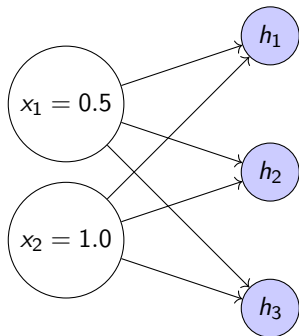


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3. Apply element-wise activation $\mathbf{a} = \text{ReLU}(\mathbf{z})$:

$$\mathbf{a} = \text{ReLU} \left(\begin{bmatrix} 0.9 \\ 0.1 \\ -0.35 \end{bmatrix} \right) = \begin{bmatrix} \max(0, 0.9) \\ \max(0, 0.1) \\ \max(0, -0.35) \end{bmatrix} = \begin{bmatrix} 0.9 \\ 0.1 \\ 0 \end{bmatrix}$$

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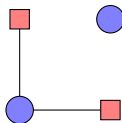
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Original Space



Cannot be separated by one line

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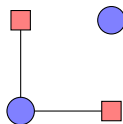
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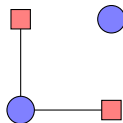
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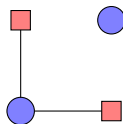
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Original Space

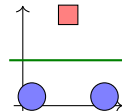


Cannot be separated by one line

Hidden Layer



Learned Representation



Now linearly separable!

The Multilayer Perceptron (MLP): Overcoming Limitations

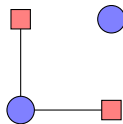
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- This allows the MLP to create complex decision boundaries.

Original Space

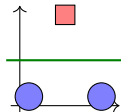


Cannot be separated by one line

Hidden Layer



Learned Representation



Now linearly separable!

Activation Functions: Introducing Non-Linearity

The Anatomy of a Neural Network

Why Do We Need Activation Functions?

The Problem: Stacking Linear Layers is Useless

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- This allows the network to learn arbitrarily complex, "wiggly" functions.

Conventional Activations: Sigmoid

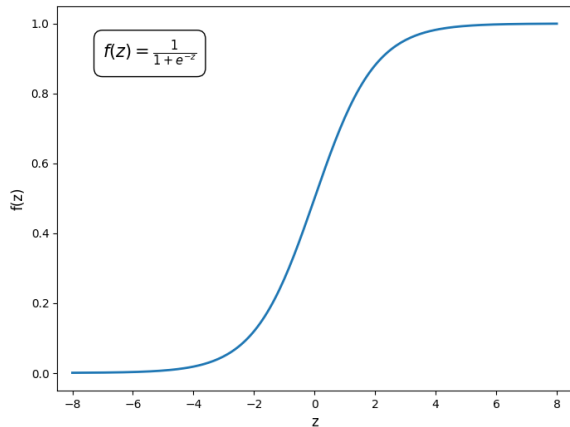


Figure: Sigmoid Function

Formula:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

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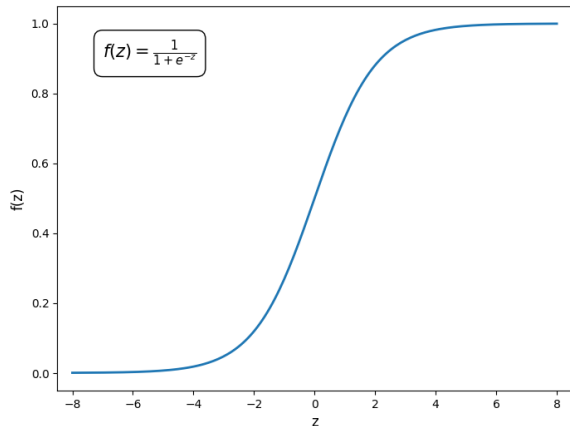


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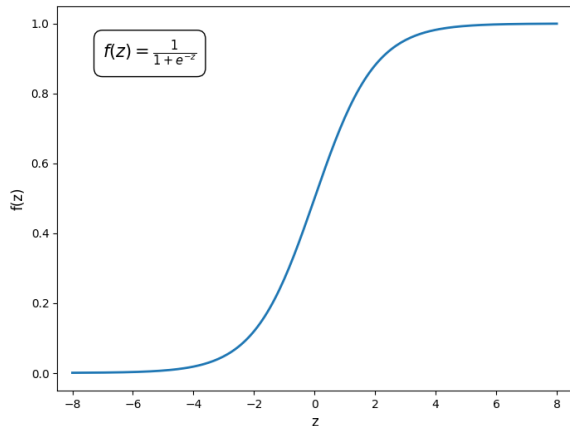


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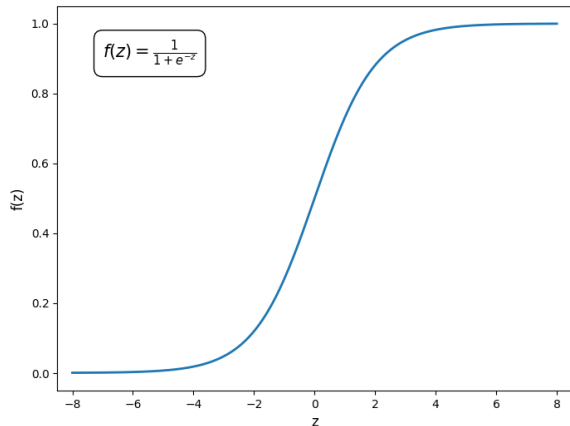


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- Useful in output layers for binary classification (predicting probabilities).

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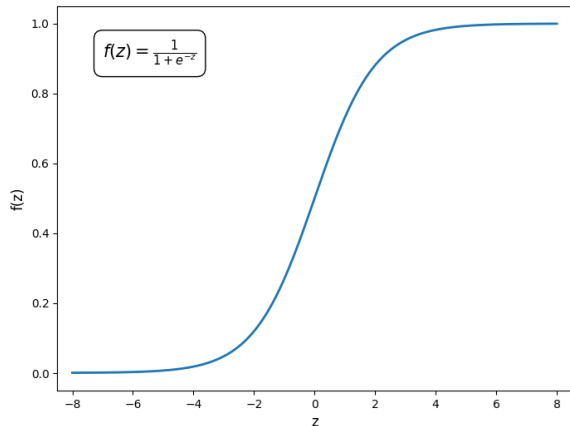


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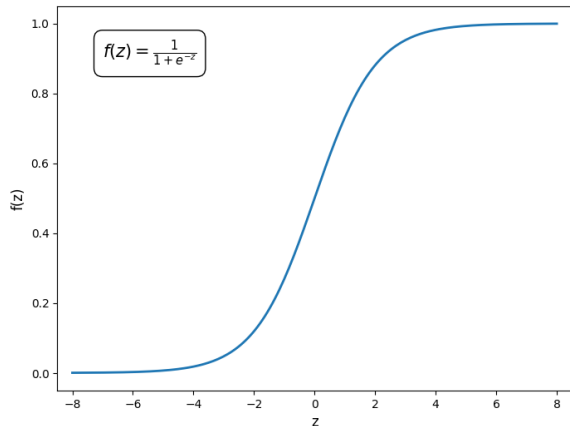


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- **Not Zero-Centered:** Outputs are always positive, which can slow down learning.

Conventional Activations: Tanh

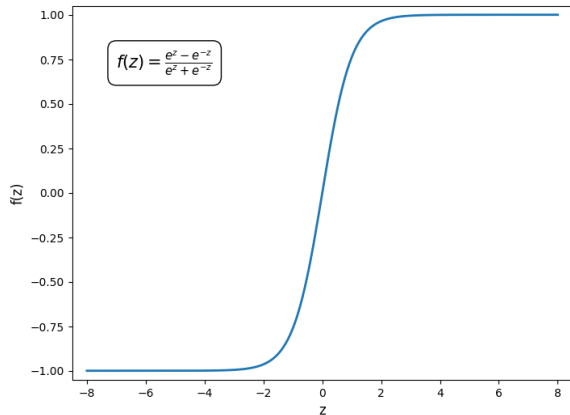


Figure: Tanh Function

Formula:

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

Conventional Activations: Tanh

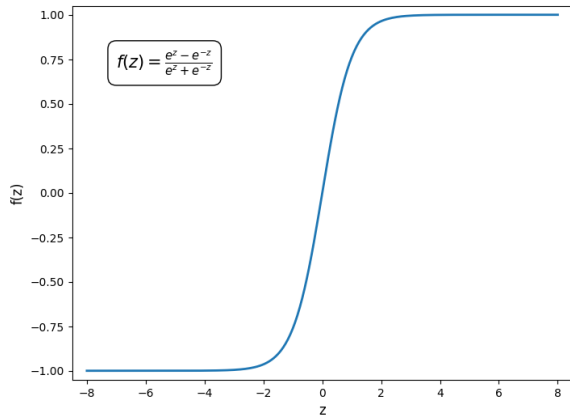


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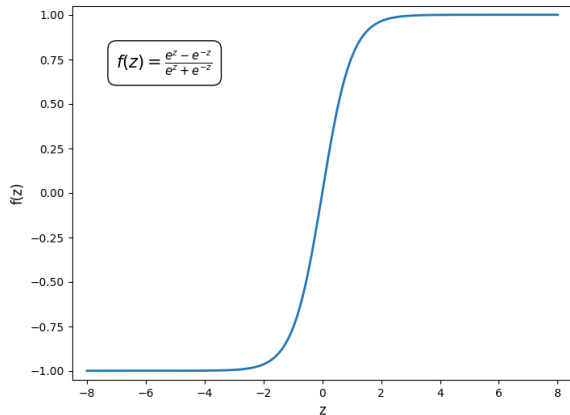


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- **Zero-Centered:** Its major advantage over Sigmoid. This property helps center the data for the next layer, often speeding up convergence.

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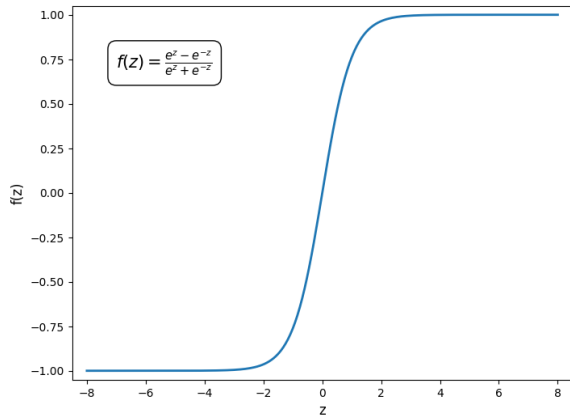


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- **Vanishing Gradients:** Like Sigmoid, the function flattens out, and its gradients disappear for large positive or negative inputs, making it difficult to use in very deep networks.

Conventional Activations: ReLU

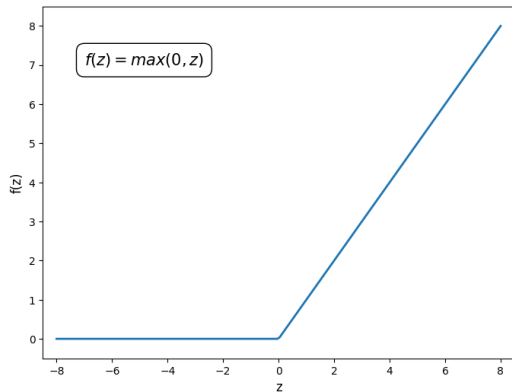


Figure: ReLU Function

Formula:

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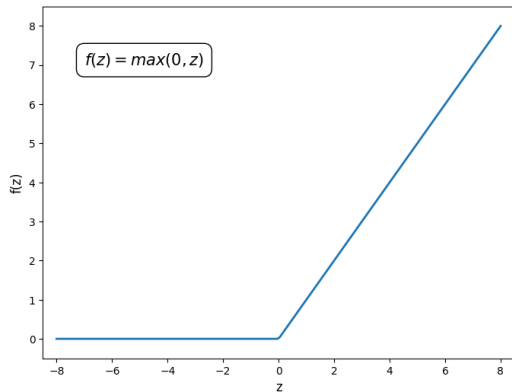


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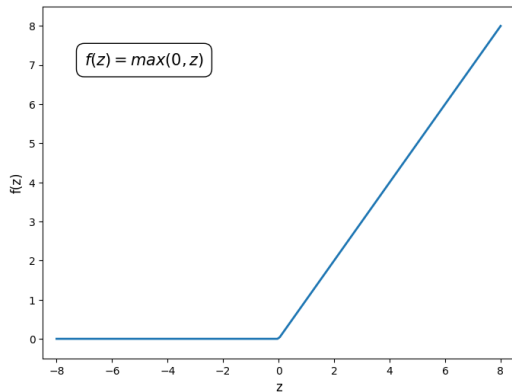


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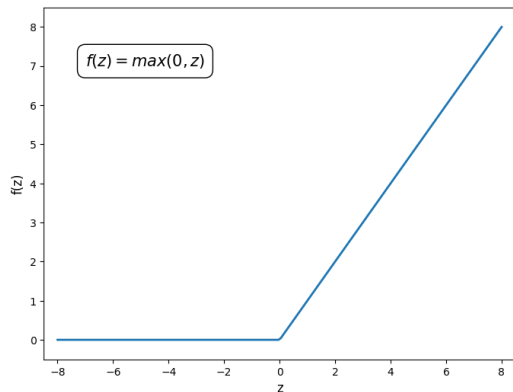


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- **Sparsity:** By outputting 0 for negative inputs, it can make the network sparse.

Conventional Activations: ReLU

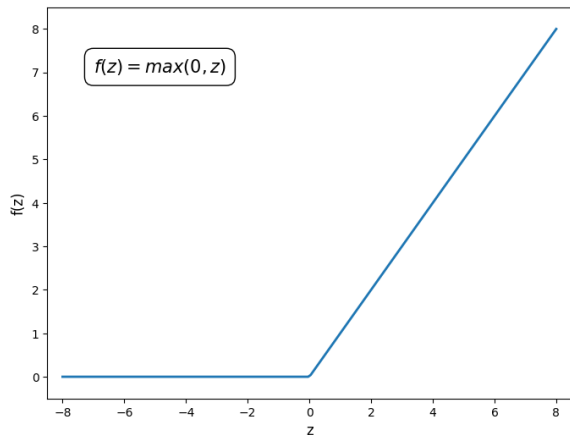


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Problems

- **The "Dying ReLU" Problem:** If a neuron's weights are updated such that its pre-activation z is always negative, it will always output 0. The gradient will also be 0, and the neuron can never recover. It effectively "dies."

The ReLU Family: Fixing the "Dying" Problem

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Leaky ReLU

$$f(z) = \begin{cases} z & \text{if } z > 0 \\ \alpha z & \text{if } z \leq 0 \end{cases}$$

(where α is a small constant like 0.01)

- **The Fix:** Introduces a small, non-zero gradient for negative inputs.
- This allows "dead" neurons to be revived. A very common and effective choice.

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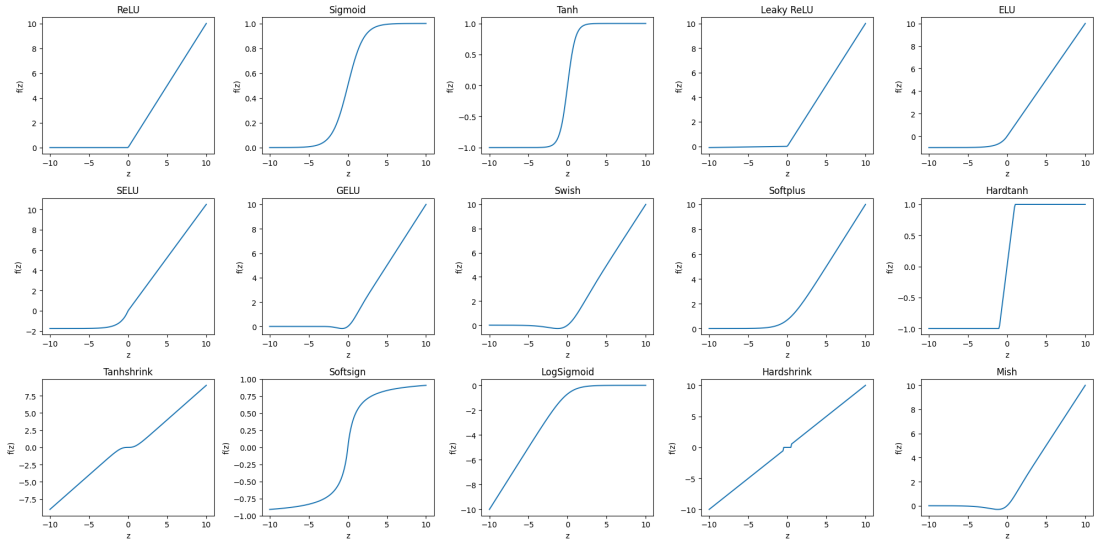
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GELU (Gaussian Error Linear Unit)

- A smoother, probabilistic alternative to ReLU.
- Became the standard in state-of-the-art **Transformer** models (e.g., GPT, BERT).

Popular Activation Functions



A Practical Guide to Choosing Activations

For Hidden Layers

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Depends on the problem's output range.

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Rule of Thumb

Never use Sigmoid or Tanh in hidden layers of modern deep networks due to the vanishing gradient problem. Their use is now almost exclusively in output layers.

The Forward Pass and Measuring Error

The Learning Loop: How Neural Networks Learn

The Forward Pass: From Input to Prediction

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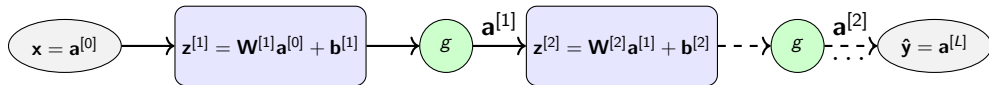


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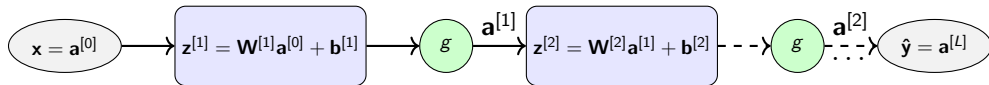


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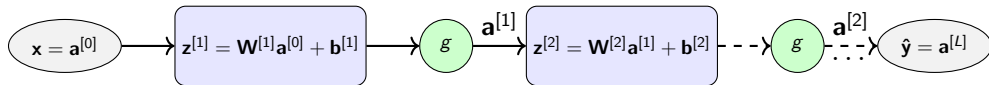


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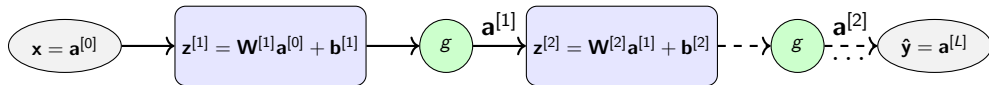


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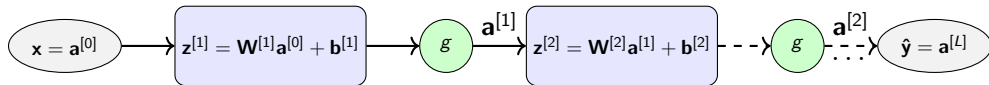


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$$\mathbf{a}^{[l]} = g(\mathbf{z}^{[l]})$$

The final activation vector $\mathbf{a}^{[L]}$ is the network's prediction $\hat{\mathbf{y}}$.

Loss Functions: Quantifying the Error

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- Calculates the average squared difference between true and predicted values.
- Penalizes large errors very heavily.

Loss Functions: Quantifying the Error

For Classification Tasks

Cross-Entropy Loss

$$\mathcal{L}_{CE} = - \sum_{c=1}^C y_c \log(\hat{y}_c)$$

- y_c is 1 if c is the true class, 0 otherwise.
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Example (3 classes):

- **True:** Cat $\rightarrow [1, 0, 0]$
- **Good Pred:** $[0.8, 0.1, 0.1] \rightarrow$ Low Loss
- **Bad Pred:** $[0.1, 0.2, 0.7] \rightarrow$ High Loss

Gradient Descent

The Learning Loop: How Neural Networks Learn

Gradient Descent: The Intuition

The Analogy: A Hiker on a Foggy Mountain

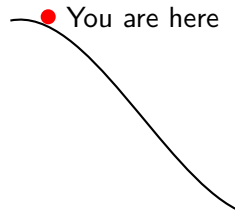
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Gradient Descent: The Intuition

The Analogy: A Hiker on a Foggy Mountain

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You ask two simple questions at every step:



Gradient Descent: The Intuition

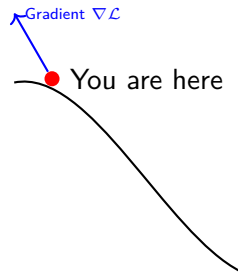
The Analogy: A Hiker on a Foggy Mountain

Imagine you are on a mountainside in thick fog and want to get to the lowest valley. What do you do?

You ask two simple questions at every step:

1. Which way is downhill?

- You feel the slope under your feet.
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- So, you take a step in the **opposite** direction.



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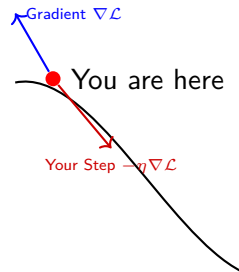
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2. How big a step should I take?

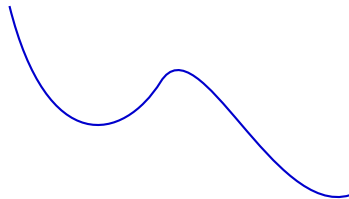
- This is your "step size" or **Learning Rate** (η).
- Too small: takes forever. Too big: overshoots the valley.



Why Gradient Descent? Navigating the Loss Landscape

The Goal

Find the parameters θ that minimize the Loss Function $\mathcal{L}(y, f(x; \theta))$. This means finding the lowest point in the landscape.



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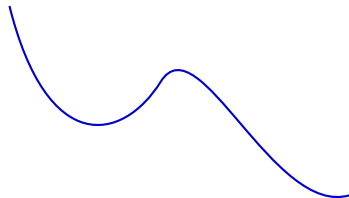
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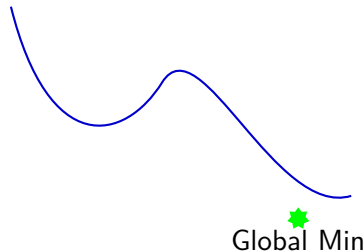
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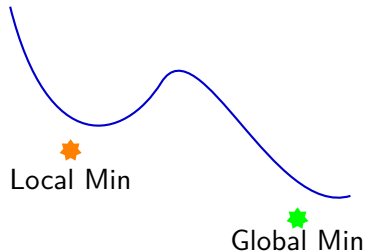
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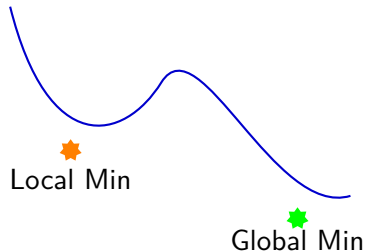
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- **Global Minimum:** The true lowest point; the optimal solution.
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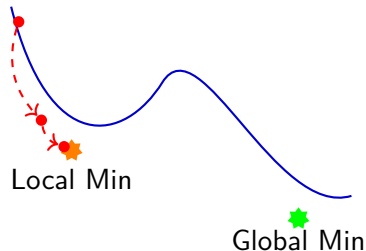
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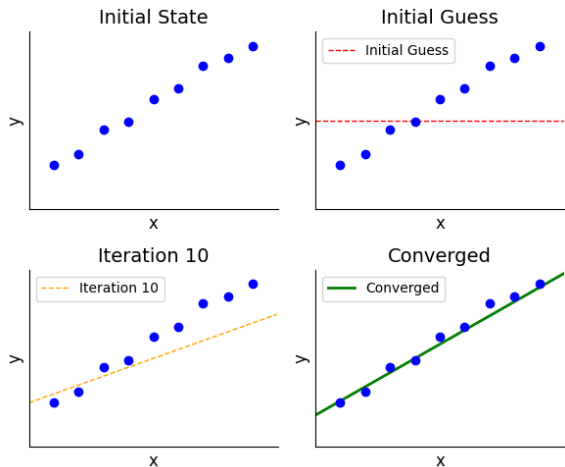
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Gradient Descent: A Practical Comparison

Aspect	Batch GD	Stochastic GD (SGD)	Mini-Batch GD
Data Usage	Entire dataset	Single data point	Mini-batch of data
Update Frequency	Once per epoch	After each data point	After each mini-batch
Convergence Speed	Slow	Fast (per iteration)	Moderate
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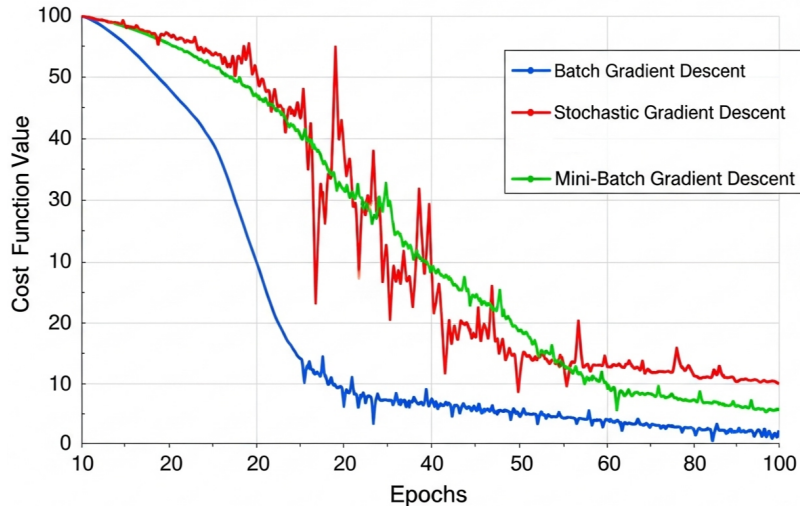
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The Winner: Mini-Batch Gradient Descent

For nearly all modern deep learning applications, **Mini-Batch GD** is the standard. It offers the best compromise between the stability of Batch GD and the speed of SGD, while being perfectly suited for parallelization on GPUs.

Convergence of Gradient Descent Methods



Backpropagation - Calculating the Gradients

The Learning Loop: How Neural Networks Learn

Backpropagation's Engine: The Chain Rule

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The Chain Rule provides the answer by multiplying local derivatives.

Imagine a function $\mathcal{L}(y(x))$. We want to find how $\mathcal{L}$ changes with respect to x , which is $\frac{d\mathcal{L}}{dx}$.

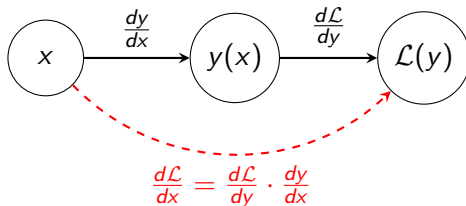
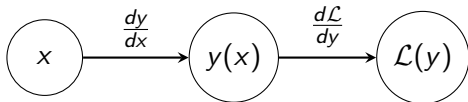
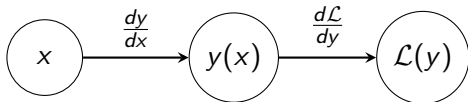


Figure: The influence of x on $\mathcal{L}$ is the product of the local influences along the path.

Scalar Case: A Simple Chain

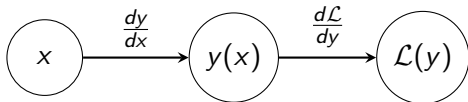


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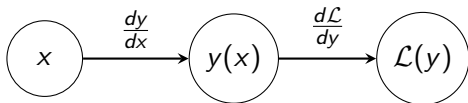
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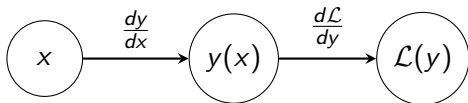


The chain rule lets us link them:

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Vector Case (Generalization)

For vector functions $\mathbf{y} = f(\mathbf{x})$ and $\mathcal{L} = g(\mathbf{y})$, the derivatives become Jacobian matrices, and the chain rule becomes matrix multiplication:

$$\nabla_{\mathbf{x}} \mathcal{L} = (\mathbf{J}_{\mathbf{y}} \mathcal{L})(\mathbf{J}_{\mathbf{x}} \mathbf{y})$$

This is the mathematical foundation for backpropagation in neural networks.

Visualizing Computations: The Computational Graph

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We can break this down into intermediate steps: $d = a + b$ and $\mathcal{L} = d \cdot c$.

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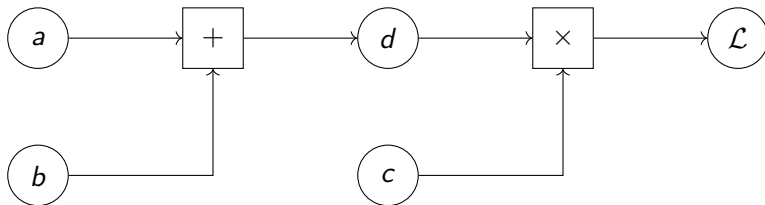


Figure: A computational graph for $\mathcal{L} = (a + b) \cdot c$. Circles are variables, boxes are operations.

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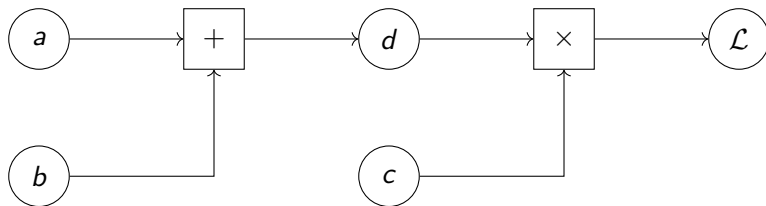


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The Two Passes of Automatic Differentiation

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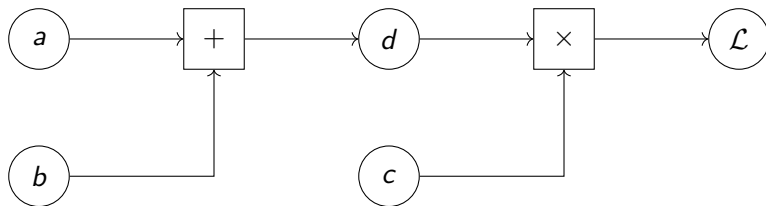


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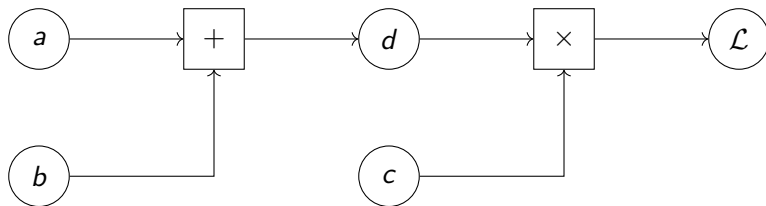


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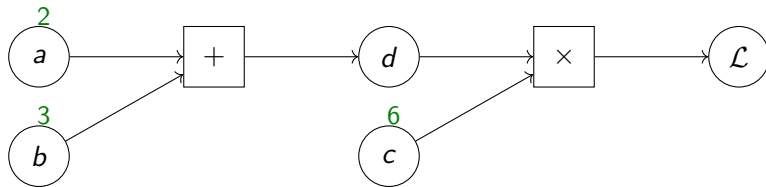
- **Forward Pass:** We start with input values (a, b, c) and compute the values of all nodes, flowing from left to right, until we get the final output $\mathcal{L}$.
- **Backward Pass (Backpropagation):** We start from the end with the gradient $\frac{\partial \mathcal{L}}{\partial \mathcal{L}} = 1$. We then go backwards (right to left), using the chain rule at each node to compute the gradient of $\mathcal{L}$ with respect to every input of that node.

Backpropagation by Hand: A Scalar Example

Let's run backpropagation on $\mathcal{L} = (a + b) \cdot c$ with inputs $a = 2, b = 3, c = 6$.

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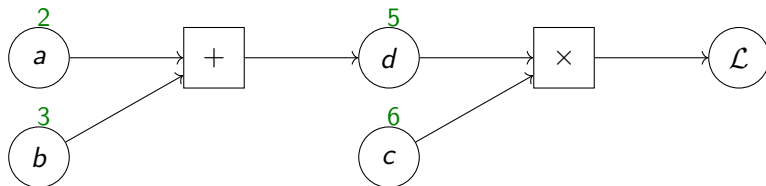


Forward Pass (Green)

Backward Pass (Red)

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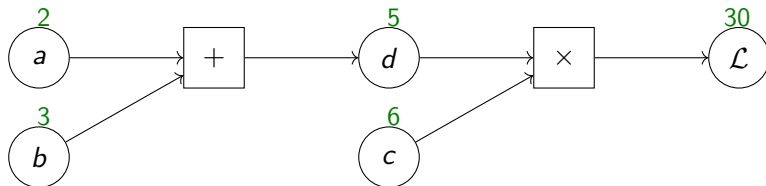
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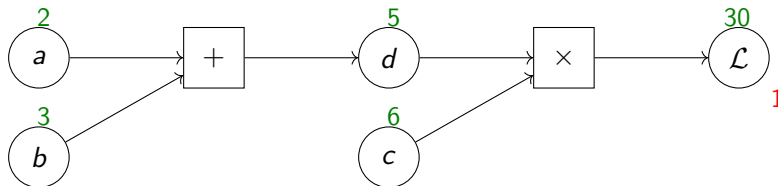
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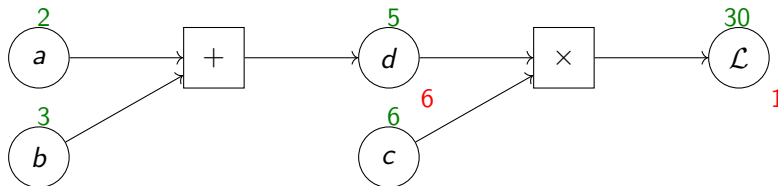
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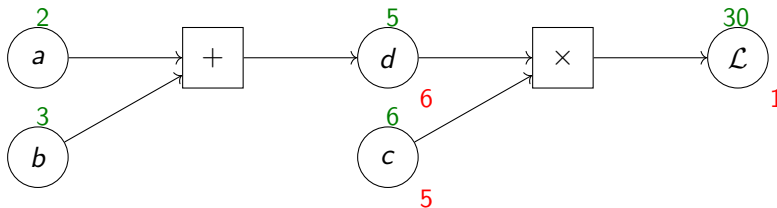
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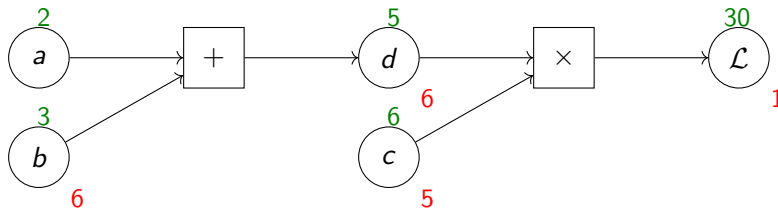
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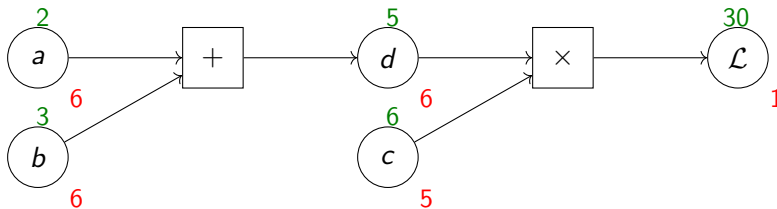
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Summary

Node	Formula	Forward Value	Local Gradient	Final Gradient
a	-	2	-	$\frac{\partial \mathcal{L}}{\partial d} \frac{\partial d}{\partial a} = 6 \cdot 1 = 6$
b	-	3	-	$\frac{\partial \mathcal{L}}{\partial d} \frac{\partial d}{\partial b} = 6 \cdot 1 = 6$
c	-	6	-	$\frac{\partial \mathcal{L}}{\partial d} \frac{\partial d}{\partial c} = 1 \cdot 5 = 5$
d	$a + b$	5	$\frac{\partial d}{\partial a} = 1, \frac{\partial d}{\partial b} = 1$	$\frac{\partial \mathcal{L}}{\partial d} \frac{\partial \mathcal{L}}{\partial c} = 1 \cdot 6 = 6$
$\mathcal{L}$	$d \cdot c$	30	$\frac{\partial \mathcal{L}}{\partial d} = c = 6, \frac{\partial \mathcal{L}}{\partial c} = d = 5$	1 (by definition)

Backpropagation in a Multilayer Perceptron

The goal is to compute the gradient of the loss $\mathcal{L}$ with respect to every weight $W_{ij}^{[l]}$ and bias $b_i^{[l]}$ in the network. We do this by propagating the error signal backwards, layer by layer.

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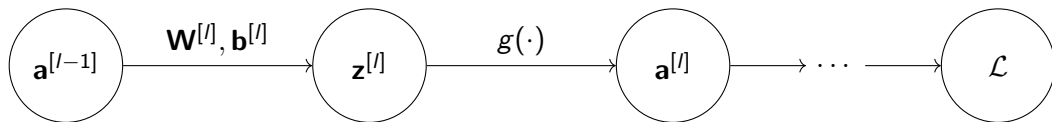


Figure: The forward pass creates a chain of dependencies. Backpropagation reverses this flow.

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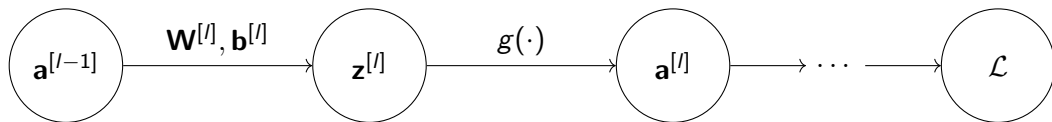


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To update the weights $\mathbf{W}^{[l]}$, we need $\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{[l]}}$.

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- ❶ **Upstream Gradient:** How the final loss changes with this layer's output. This is passed back from layer $l + 1$.

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$$\underbrace{\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{[l]}}}_{\text{Our Goal}} = \underbrace{\frac{\partial \mathcal{L}}{\partial \mathbf{a}^{[l]}}}_{\text{1. Upstream Gradient}} \cdot \underbrace{\frac{\partial \mathbf{a}^{[l]}}{\partial \mathbf{z}^{[l]}}}_{\text{2. Activation Gradient}} \cdot \underbrace{\frac{\partial \mathbf{z}^{[l]}}{\partial \mathbf{W}^{[l]}}}_{\text{3. Local Gradient}}$$

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Backpropagation in a Multilayer Perceptron

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The Weight Update Rules: The Full Derivation

Let's derive the update rules for the weights $\mathbf{W}^{[l]}$ and biases $\mathbf{b}^{[l]}$ of a hidden layer l .

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For the output layer L , this is simpler: $\delta^{[L]} = \nabla_{\mathbf{a}^{[L]}} \mathcal{L} \odot g'(\mathbf{z}^{[L]})$.

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These gradients are then used in an optimizer (like SGD) to update the parameters.

The XOR Problem

The Learning Loop: How Neural Networks Learn

The XOR Problem: A Classic Challenge

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XOR Truth Table

Input A	Input B	Output
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1	0	1
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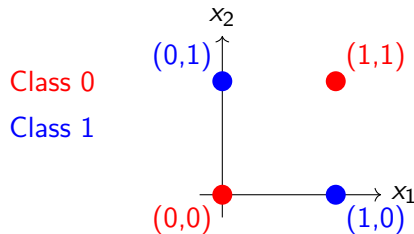
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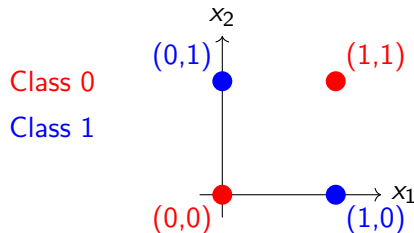
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Key Insight

No single straight line can separate the red points (Class 0) from the blue points (Class 1). This is why simple linear models like the Perceptron fail.

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The equation of a line in 2D, or a hyperplane in higher dimensions, which separates the two predicted classes is:

$$w_1x_1 + w_2x_2 + b = 0$$

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Substituting these into (4):

$$\underbrace{w_1}_{\approx 1} + \underbrace{w_2}_{\approx 1} + \underbrace{b}_{\approx 0} \approx 0 \implies 1 + 1 + 0 \approx 0 \implies \mathbf{2 \approx 0}$$

This is a contradiction, proving no linear model can solve XOR.

The Solution: A Multi-Layer Perceptron (MLP)

To solve XOR, we need a hidden layer with a **non-linear activation function**. This allows the network to create non-linear decision boundaries.

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The 2-2-1 Architecture

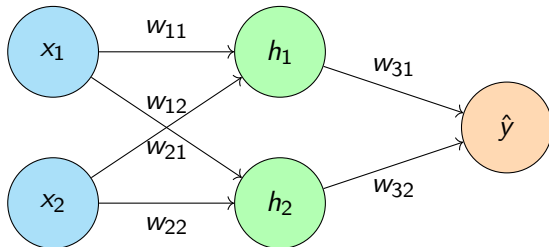
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Hidden Layer Computations

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$$L(y, \hat{y}) = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

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Goal

Calculate the network's prediction $\hat{y}$ and the resulting loss L for input $(1, 0)$.

Forward Pass: Step 1 (Hidden Layer)

Input: $x_1 = 1, x_2 = 0$. **Target:** $y = 1$.

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- For hidden neuron h_1 :

$$\begin{aligned} z_1 &= w_{11}x_1 + w_{21}x_2 + b_1 \\ &= (1.0 \times 1) + (1.0 \times 0) + 0.0 = \mathbf{1.0} \end{aligned}$$

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- For hidden neuron h_2 :

$$\begin{aligned} z_2 &= w_{12}x_1 + w_{22}x_2 + b_2 \\ &= (-1.0 \times 1) + (-1.0 \times 0) + 1.0 = \mathbf{0.0} \end{aligned}$$

Forward Pass: Step 2 (Hidden Layer)

Step 2: Apply Sigmoid Activation

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- $h_1 = \sigma(z_1) = \sigma(1.0) = \frac{1}{1+e^{-1}} \approx \mathbf{0.731}$
- $h_2 = \sigma(z_2) = \sigma(0.0) = \frac{1}{1+e^0} = \mathbf{0.500}$

Forward Pass: Step 3, 4 (Output)

Hidden Activations: $h_1 = 0.731$, $h_2 = 0.500$. **Target:** $y = 1$.

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$$\begin{aligned} z_3 &= w_{31}h_1 + w_{32}h_2 + b_3 \\ &= (2.0 \times 0.731) + (-1.0 \times 0.500) + (-1.0) \\ &= 1.462 - 0.500 - 1.000 = -\mathbf{0.038} \end{aligned}$$

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$$\hat{y} = \sigma(z_3) = \sigma(-0.038) = \frac{1}{1 + e^{0.038}} \approx \mathbf{0.491}$$

Forward Pass: Step 5 (Loss)

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Step 5: Calculate Loss ($\mathcal{L}$)

$$\begin{aligned}\mathcal{L} &= -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \\ &= -[1 \times \log(0.491) + (0) \times \log(1 - 0.491)] \\ &= -(-0.712) = \mathbf{0.712}\end{aligned}$$

Backpropagation: Mathematical Derivation

Goal

Find $\frac{\partial \mathcal{L}}{\partial w_{ij}}$ for all weights w_{ij} to update them using gradient descent:

$$w_{new} = w_{old} - \alpha \frac{\partial \mathcal{L}}{\partial w_{old}}$$

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Key Derivatives

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 $\frac{\partial \mathcal{L}}{\partial \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})}$

Chain Rule Structure We calculate gradients "backwards" from the output layer.

- 1 Calculate gradients for Output Layer weights (W_2, b_2).
- 2 Propagate error back to the Hidden Layer.
- 3 Calculate gradients for Hidden Layer weights (W_1, b_1).

Derivation: Layer-by-Layer Gradient Flow (Output)

Gradient w.r.t. Output Pre-Activation (z_3)

Derivation: Layer-by-Layer Gradient Flow (Output)

Gradient w.r.t. Output Pre-Activation (z_3)

Using the chain rule: $\frac{\partial L}{\partial z_3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_3}$

Derivation: Layer-by-Layer Gradient Flow (Output)

Gradient w.r.t. Output Pre-Activation (z_3)

Using the chain rule: $\frac{\partial L}{\partial z_3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_3}$

$$\begin{aligned}\frac{\partial L}{\partial z_3} &= \underbrace{\left(\frac{\hat{y} - y}{\hat{y}(1 - \hat{y})} \right)}_{\partial L / \partial \hat{y}} \cdot \underbrace{(\hat{y}(1 - \hat{y}))}_{\partial \hat{y} / \partial z_3} \\ &= \hat{y} - y \quad (\text{a very convenient simplification!})\end{aligned}$$

Let's call this error term $\delta_3 = \hat{y} - y$.

Derivation: Layer-by-Layer Gradient Flow (Output)

Gradients for Output Layer Weights (w_{31} , w_{32} , b_3)

Derivation: Layer-by-Layer Gradient Flow (Output)

Gradients for Output Layer Weights (w_{31} , w_{32} , b_3)

Apply the chain rule again:

- $\frac{\partial \mathcal{L}}{\partial w_{31}} = \frac{\partial \mathcal{L}}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_{31}} = \delta_3 \cdot h_1$

Derivation: Layer-by-Layer Gradient Flow (Output)

Gradients for Output Layer Weights (w_{31}, w_{32}, b_3)

Apply the chain rule again:

- $\frac{\partial \mathcal{L}}{\partial w_{31}} = \frac{\partial \mathcal{L}}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_{31}} = \delta_3 \cdot h_1$
- $\frac{\partial \mathcal{L}}{\partial w_{32}} = \frac{\partial \mathcal{L}}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_{32}} = \delta_3 \cdot h_2$

Derivation: Layer-by-Layer Gradient Flow (Output)

Gradients for Output Layer Weights (w_{31} , w_{32} , b_3)

Apply the chain rule again:

- $\frac{\partial \mathcal{L}}{\partial w_{31}} = \frac{\partial \mathcal{L}}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_{31}} = \delta_3 \cdot h_1$
- $\frac{\partial \mathcal{L}}{\partial w_{32}} = \frac{\partial \mathcal{L}}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_{32}} = \delta_3 \cdot h_2$
- $\frac{\partial \mathcal{L}}{\partial b_3} = \frac{\partial \mathcal{L}}{\partial z_3} \cdot \frac{\partial z_3}{\partial b_3} = \delta_3 \cdot 1 = \delta_3$

Derivation: Layer-by-Layer Gradient Flow (Hidden)

Propagating the Error to the Hidden Layer (z_1, z_2)

Derivation: Layer-by-Layer Gradient Flow (Hidden)

Propagating the Error to the Hidden Layer (z_1, z_2)

We need $\frac{\partial \mathcal{L}}{\partial z_1}$ and $\frac{\partial \mathcal{L}}{\partial z_2}$.

Derivation: Layer-by-Layer Gradient Flow (Hidden)

Propagating the Error to the Hidden Layer (z_1, z_2)

We need $\frac{\partial \mathcal{L}}{\partial z_1}$ and $\frac{\partial \mathcal{L}}{\partial z_2}$.

$$\frac{\partial \mathcal{L}}{\partial z_1} = \frac{\partial \mathcal{L}}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} = \delta_3 \cdot w_{31} \cdot \sigma'(z_1)$$

$$\frac{\partial \mathcal{L}}{\partial z_2} = \frac{\partial \mathcal{L}}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_2} \cdot \frac{\partial h_2}{\partial z_2} = \delta_3 \cdot w_{32} \cdot \sigma'(z_2)$$

Derivation: Layer-by-Layer Gradient Flow (Hidden)

Propagating the Error to the Hidden Layer (z_1, z_2)

We need $\frac{\partial \mathcal{L}}{\partial z_1}$ and $\frac{\partial \mathcal{L}}{\partial z_2}$.

$$\frac{\partial \mathcal{L}}{\partial z_1} = \frac{\partial \mathcal{L}}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} = \delta_3 \cdot w_{31} \cdot \sigma'(z_1)$$

$$\frac{\partial \mathcal{L}}{\partial z_2} = \frac{\partial \mathcal{L}}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_2} \cdot \frac{\partial h_2}{\partial z_2} = \delta_3 \cdot w_{32} \cdot \sigma'(z_2)$$

We can define the hidden layer error terms $\delta_1 = \frac{\partial \mathcal{L}}{\partial z_1}$ and $\delta_2 = \frac{\partial \mathcal{L}}{\partial z_2}$.

Derivation: Layer-by-Layer Gradient Flow (Hidden)

Gradients for Hidden Layer Weights (w_{11} , w_{21} , etc.)

Derivation: Layer-by-Layer Gradient Flow (Hidden)

Gradients for Hidden Layer Weights (w_{11} , w_{21} , etc.)

Apply the chain rule one last time:

- $$\frac{\partial \mathcal{L}}{\partial w_{11}} = \frac{\partial \mathcal{L}}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{11}} = \delta_1 \cdot x_1$$

Derivation: Layer-by-Layer Gradient Flow (Hidden)

Gradients for Hidden Layer Weights (w_{11} , w_{21} , etc.)

Apply the chain rule one last time:

- $\frac{\partial \mathcal{L}}{\partial w_{11}} = \frac{\partial \mathcal{L}}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{11}} = \delta_1 \cdot x_1$
- $\frac{\partial \mathcal{L}}{\partial w_{21}} = \frac{\partial \mathcal{L}}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{21}} = \delta_1 \cdot x_2$

Derivation: Layer-by-Layer Gradient Flow (Hidden)

Gradients for Hidden Layer Weights (w_{11} , w_{21} , etc.)

Apply the chain rule one last time:

- $\frac{\partial \mathcal{L}}{\partial w_{11}} = \frac{\partial \mathcal{L}}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{11}} = \delta_1 \cdot x_1$
- $\frac{\partial \mathcal{L}}{\partial w_{21}} = \frac{\partial \mathcal{L}}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{21}} = \delta_1 \cdot x_2$
- $\frac{\partial \mathcal{L}}{\partial b_1} = \frac{\partial \mathcal{L}}{\partial z_1} \cdot \frac{\partial z_1}{\partial b_1} = \delta_1$

(The same pattern applies for weights connected to h_2 , using δ_2).

Backward Pass: Hand Calculation (Step 1 - Output)

Given from Forward Pass: $\hat{y} = 0.491, y = 1, h_1 = 0.731, h_2 = 0.500, x_1 = 1, x_2 = 0$.

Backward Pass: Hand Calculation (Step 1 - Output)

Given from Forward Pass: $\hat{y} = 0.491, y = 1, h_1 = 0.731, h_2 = 0.500, x_1 = 1, x_2 = 0$.

Step 1: Output Layer Gradients

- Calculate the error term δ_3 :

$$\delta_3 = \frac{\partial \mathcal{L}}{\partial z_3} = \hat{y} - y = 0.491 - 1 = -\mathbf{0.509}$$

Backward Pass: Hand Calculation (Step 1 - Output)

Given from Forward Pass: $\hat{y} = 0.491, y = 1, h_1 = 0.731, h_2 = 0.500, x_1 = 1, x_2 = 0$.

Step 1: Output Layer Gradients

- Calculate the error term δ_3 :

$$\delta_3 = \frac{\partial \mathcal{L}}{\partial z_3} = \hat{y} - y = 0.491 - 1 = -\mathbf{0.509}$$

- Calculate gradients for weights:

$$\frac{\partial \mathcal{L}}{\partial w_{31}} = \delta_3 \cdot h_1 = (-0.509) \times 0.731 = -\mathbf{0.372}$$

$$\frac{\partial \mathcal{L}}{\partial w_{32}} = \delta_3 \cdot h_2 = (-0.509) \times 0.500 = -\mathbf{0.255}$$

Backward Pass: Hand Calculation (Step 1 - Output)

Given from Forward Pass: $\hat{y} = 0.491, y = 1, h_1 = 0.731, h_2 = 0.500, x_1 = 1, x_2 = 0$.

Step 1 Continued: Output Layer Gradients

- Calculate gradient for bias:

$$\frac{\partial \mathcal{L}}{\partial b_3} = \delta_3 = -\mathbf{0.509}$$

Backward Pass: Hand Calculation (Step 2 - Hidden)

Given from Previous Step: $\delta_3 = -0.509$, $w_{31} = 2.0$, $w_{32} = -1.0$. **Given from Forward Pass:** $h_1 = 0.731$, $h_2 = 0.500$, $x_1 = 1$, $x_2 = 0$.

Backward Pass: Hand Calculation (Step 2 - Hidden)

Given from Previous Step: $\delta_3 = -0.509$, $w_{31} = 2.0$, $w_{32} = -1.0$. **Given from Forward Pass:** $h_1 = 0.731$, $h_2 = 0.500$, $x_1 = 1$, $x_2 = 0$.

Step 2: Hidden Layer Gradients

Backward Pass: Hand Calculation (Step 2 - Hidden)

Given from Previous Step: $\delta_3 = -0.509$, $w_{31} = 2.0$, $w_{32} = -1.0$. **Given from Forward Pass:** $h_1 = 0.731$, $h_2 = 0.500$, $x_1 = 1$, $x_2 = 0$.

Step 2: Hidden Layer Gradients

- First, calculate the hidden layer error terms δ_1, δ_2 :

$$\begin{aligned}\delta_1 &= \frac{\partial \mathcal{L}}{\partial z_1} = (\delta_3 \cdot w_{31}) \cdot \sigma'(z_1) \\ &= (-0.509 \times 2.0) \cdot (h_1(1 - h_1)) \\ &= (-1.018) \cdot (0.731 \times (1 - 0.731)) = \mathbf{-0.200}\end{aligned}$$

$$\begin{aligned}\delta_2 &= \frac{\partial \mathcal{L}}{\partial z_2} = (\delta_3 \cdot w_{32}) \cdot \sigma'(z_2) \\ &= (-0.509 \times -1.0) \cdot (h_2(1 - h_2)) \\ &= (0.509) \cdot (0.5 \times (1 - 0.5)) = \mathbf{0.127}\end{aligned}$$

Backward Pass: Hand Calculation (Step 2 Cont.)

Given: $\delta_1 = -0.200$, $\delta_2 = 0.127$, $x_1 = 1$, $x_2 = 0$.

Step 2 Continued: Gradients for Hidden Weights & Biases

Backward Pass: Hand Calculation (Step 2 Cont.)

Given: $\delta_1 = -0.200$, $\delta_2 = 0.127$, $x_1 = 1$, $x_2 = 0$.

Step 2 Continued: Gradients for Hidden Weights & Biases

- Weights connected to h_1 :

$$\frac{\partial \mathcal{L}}{\partial w_{11}} = \delta_1 \cdot x_1 = -0.200 \times 1 = -\mathbf{0.200} \qquad \frac{\partial L}{\partial w_{21}} = \delta_1 \cdot x_2 = -0.200 \times 0 = \mathbf{0.0}$$

Backward Pass: Hand Calculation (Step 2 Cont.)

Given: $\delta_1 = -0.200$, $\delta_2 = 0.127$, $x_1 = 1$, $x_2 = 0$.

Step 2 Continued: Gradients for Hidden Weights & Biases

- Weights connected to h_1 :

$$\frac{\partial \mathcal{L}}{\partial w_{11}} = \delta_1 \cdot x_1 = -0.200 \times 1 = \mathbf{-0.200}$$

$$\frac{\partial L}{\partial w_{21}} = \delta_1 \cdot x_2 = -0.200 \times 0 = \mathbf{0.0}$$

- Weights connected to h_2 :

$$\frac{\partial \mathcal{L}}{\partial w_{12}} = \delta_2 \cdot x_1 = 0.127 \times 1 = \mathbf{0.127}$$

$$\frac{\partial L}{\partial w_{22}} = \delta_2 \cdot x_2 = 0.127 \times 0 = \mathbf{0.0}$$

Backward Pass: Hand Calculation (Step 2 Cont.)

Given: $\delta_1 = -0.200$, $\delta_2 = 0.127$, $x_1 = 1$, $x_2 = 0$.

Step 2 Continued: Gradients for Hidden Weights & Biases

- Biases:

$$\frac{\partial \mathcal{L}}{\partial b_1} = \delta_1 = -\mathbf{0.200}$$

$$\frac{\partial \mathcal{L}}{\partial b_2} = \delta_2 = \mathbf{0.127}$$

Weight Updates

Now we use the calculated gradients to update the weights using Gradient Descent.

Update Rule

$$w_{new} = w_{old} - \alpha \frac{\partial \mathcal{L}}{\partial w} \quad (\text{where } \alpha \text{ is the learning rate})$$

Weight Updates

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Let's use a learning rate $\alpha = 0.1$.

- **Output Layer Updates:**

- $w_{31} = 2.0 - 0.1 \times (-0.372) = \mathbf{2.0372}$
- $w_{32} = -1.0 - 0.1 \times (-0.255) = \mathbf{-0.9745}$
- $b_3 = -1.0 - 0.1 \times (-0.509) = \mathbf{-0.9491}$

Weight Updates

Now we use the calculated gradients to update the weights using Gradient Descent.

Update Rule

$$w_{new} = w_{old} - \alpha \frac{\partial \mathcal{L}}{\partial w} \quad (\text{where } \alpha \text{ is the learning rate})$$

Let's use a learning rate $\alpha = 0.1$.

- **Output Layer Updates:**

- $w_{31} = 2.0 - 0.1 \times (-0.372) = \mathbf{2.0372}$
- $w_{32} = -1.0 - 0.1 \times (-0.255) = \mathbf{-0.9745}$
- $b_3 = -1.0 - 0.1 \times (-0.509) = \mathbf{-0.9491}$

- **Hidden Layer Updates:**

- $w_{11} = 1.0 - 0.1 \times (-0.200) = \mathbf{1.020}$
- $w_{12} = -1.0 - 0.1 \times (0.127) = \mathbf{-1.0127}$
- $b_1 = 0.0 - 0.1 \times (-0.200) = \mathbf{0.020}$
- $b_2 = 1.0 - 0.1 \times (0.127) = \mathbf{0.9873}$

The weights w_{21} and w_{22} remain unchanged because their gradients were zero (since $x_2 = 0$).

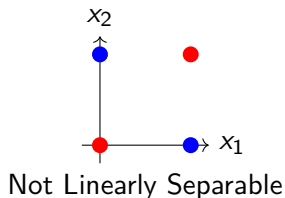
Why Nonlinear Activation Succeeds: Feature Transformation

The hidden layer's crucial role is to **transform the input data** into a new representation where the problem becomes linearly separable.

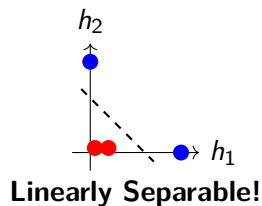
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Original Input Space



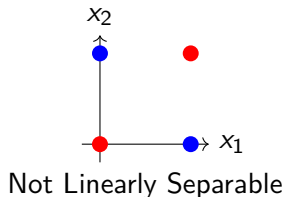
Hidden Layer Feature Space



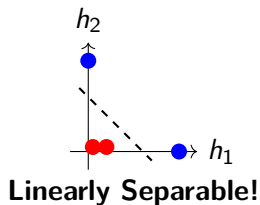
Why Nonlinear Activation Succeeds: Feature Transformation

The hidden layer's crucial role is to **transform the input data** into a new representation where the problem becomes linearly separable.

Original Input Space



Hidden Layer Feature Space



The "Magic" of Hidden Layers

The hidden neurons learn functions that act as feature detectors - one neuron learns "OR", another learns "NAND".